Data Permissions, Sources, Tools, Connections

Complete Course for Enterprise Data Professionals

Shivgan Joshi

Udemy

2026

Course Overview

What You Will Learn

- Working with Data at enterprise scale (20-100M+ rows)
- Understanding Permissions and data governance
- Exploring data sources across different environments
- Tools like Toad, Teradata, PySpark, SAS
- Connections and troubleshooting real errors

Prerequisite

Basic SQL knowledge recommended.

Course Complete Reference Table

Sec	Topic	Key Concepts	Tools / Commands
1	Data Basics	Permissions, Big Data (20-100M rows), Hive tables	SELECT, FROM, WHERE, LIMIT
2	Permissions	PROD/DEV/UAT, Table not found, Confidential levels	Wiki search, Ticketing system
3	ODBC	Manual DSN, Auto connection strings	Driver={Teradata}; DBCName=tdprod
3	Toad	ALT+F+N, Native vs ODBC, New Connection Wizard	Teradata(.NET), Oracle(ODP.NET), Vertica(ODBC only)
3	Auth	Manual, Windows SSO (Teradata/Oracle), Extra Strings	Mechanism, Account String, LDAP, Kerberos
4	Teradata	MPP, Big Data #1, High cost, EDW	SELECT TOP 100, Primary Index
4	Oracle	OLTP, Shared Everything, Very High cost	PL/SQL, TNS, Wallet
4	Vertica	Columnar, Fast analytics, 64-bit ODBC only	Columnar storage, Projections
4	Hive	Data Lake, Cheap storage, Batch only	HDFS, Parquet, ORC
5	Banking	Teradata=Truth, Vertica=Speed, Hive=Scale	10yr history, 50M txns/day, 500B archive
5	Joins	INNER, LEFT, RIGHT, FULL, CROSS, SELF, ANTI, SEMI	Venn diagram
5	Optimization	Indexes, Filter early, Avoid SELECT *, EXPLAIN	Partition pruning, LIMIT
5	AI	SQL to PySpark conversion	df.groupBy().sum().show()
6	SAS	Logistic Regression, Regulatory compliance	PROC LOGISTIC, PROC PRINT
Bank Golden Rule: Teradata=Truth | Vertica=Speed | Hive=Scale			
Do not join across systems in real-time. Move data to one place first.			

Total: 48 Slides | 20+ Lectures | Real-life Scenarios

Course Format

Example

Course Structure

- 20+ Lectures
- Real-life Issue Scenarios
- Hands-on Examples
- Quiz Questions
- Practice Exercises

Total Slides: 48

Lecture 1: Introduction to Working with Data

Definition

Data Permissions Checking permissions helps understand:

- What the data is about
- Who can access it
- What operations are allowed

Example

Data exists in enterprise at different levels, with different permissions, and in various formats.

Lecture 2: Data, Big Data, and Very Big Data

Scale Matters

Most of our work will with very big data:

- 20 - 100 million data points
- Resides in Hive tables
- Accessed through various challenges

Data Size Comparison Table

Size Category	Rows	Typical Location
Small	Under 1M	Excel, CSV, Access
Big	1M - 20M	SQL Server, Oracle, MySQL
Very Big	20M - 100M+	Hive, Teradata, Big Data

Key Insight

Very big data resides in Hive tables but accessed through various challenges.

Lecture 3: Implementation Team and Framework

How the Implementation Team Works

- **Who they are:** Technical team (DBAs, data engineers, platform leads)
- **What they do:** Deploy, schedule, and monitor data models in production
- **Why it's complex:** Models run across different databases (Teradata, Vertica, Hive)
- **Key reality:** Models might need high resources (CPU, memory, I/O)

Framework in Action

1. Business defines model logic
2. Implementation team translates to SQL / PySpark / SAS
3. Team requests permissions and connections
4. Execution happens in batch or near-real-time
5. Team monitors performance and troubleshoots errors

Example

Lecture 3: Implementation Team and Framework

- Implementation is a complex phenomenon
- The implementation team:
  - Runs the models
  - Requires high resources
  - Manages technical infrastructure

Models might need high resources - plan ahead!

1. Connect to the database (ODBC or Native)
2. Check your permissions
3. Write your query (SELECT, FROM, WHERE)
4. Execute and fetch results
5. Handle errors if any

Example Query:

SELECT * FROM employee_data
WHERE department = SALES
LIMIT 100;

Where Data Resides in Enterprise

Hive / Teradata →

SQL Server / Oracle →

Excel / CSV Files →

Enterprise Data Lake

Lecture 5: Raising Permissions Steps

Permission Request Process

1. You get a new error (access denied, table not found)
2. Search for Notes or Wiki about the permission
3. Identify the required permission level
4. Submit a request via ticketing system

Example

Example: You have PROD and DEV permission, but not UAT permission.

Lecture 6: Permission Errors

Common Permission Errors

- **Table not found** - (You don't have SELECT permission)
- **Database not found** - (No access to the schema)
- **Server connection refused** - (Network or firewall)
- **ODBC driver error** - (Driver not installed or configured)

Permission Error Resolution Matrix

Error Message	**Solution**
Table not found	Request SELECT permission from DBA
Database not found	Request schema access
Access denied	Check role or group membership
View not found	Missing underlying table access

Lecture 7: Confidential vs Very Confidential Permission

Normal Permission

- PII data
- Financial records
- Internal reports
- Accessible by many teams

Very Confidential

- Trade secrets
- Executive data
- Merger info
- Needs special approval

Permission Levels by Data Type

Data Type	Permission Level	Approval
Public Data	Low	Self-service
Internal Reports	Medium	Manager
PII / PHI	High	Compliance
Trade Secrets	Very High	Executive

Best Practices for Permissions

Security Best Practices

- Least privilege principle
- Regular permission audits
- Use groups instead of individual grants
- Document all permission requests
- Set expiration for temporary access

Always verify permission level before accessing sensitive data!

Definition

ODBC Open Database Connectivity - Standard API for accessing database management systems.

- **Manual ODBC**: Configure DSN manually in Control Panel
- **Automatic ODBC**: Connection strings or application config

ODBC Connection String Examples

Teradata Example:

Driver={Teradata};DBCName=tdprod;
Database=my_db;UID=user;PWD=pass;

Oracle Example:

Driver={Oracle in OraClient};
Data Source=ORCL;UID=user;PWD=pass;

Lecture 9: Connections at TOAD

TOAD Connection Screens

- **New Connection Wizard**
- Connection Type selection (Oracle, SQL Server, Teradata)
- Authentication method (Windows, LDAP, Database)
- Advanced settings (SSL, Proxy)

Shortcut: ALT+F+N

Toad Connection Methods: Native vs ODBC

Feature	**Teradata**	**Oracle**	**Vertica**
Native Driver	.NET Provider	ODP.NET	None
ODBC Support	Yes	Yes	Only
Windows SSO	Yes	Yes	No
LDAP Support	Yes	Yes	No
Recommended	**Native**	**Native**	**ODBC**

Vertica ODBC Requirements

Critical: Bitness Mismatch

- Toad 2025 R2+ is 64-bit only
- Must use 64-bit Vertica ODBC driver
- 32-bit driver will fail to connect

Example

Solution Download and install 64-bit Vertica ODBC driver from vendor website.

Authentication Methods Comparison

Manual

- Type user/password
- Works for ALL
- Most common

Windows SSO

- Uses Windows login
- No password typing
- Not for Vertica

Extra Strings

- Mechanism
- Account String
- LDAP or Kerberos

Database Authentication by Type

Database	Manual	Windows SSO	Extra Strings
Teradata	Yes	Yes	Mechanism, Account
Oracle	Yes	Yes	TNS, LDAP, Wallet
Vertica	Yes	No	DSN Parameters
SQL Server	Yes	Yes	None needed
MySQL	Yes	No	None needed

Lecture 10: Different Kinds of Databases

Database Type	**Examples**
Relational	Oracle, MySQL, SQL Server, PostgreSQL
MPP / Data Warehouse	Teradata, Redshift, Snowflake, Vertica
Hadoop / Big Data	Hive, HBase, Spark SQL
NoSQL	MongoDB, Cassandra
Columnar	ClickHouse (Vertica listed above as MPP)

Note: Vertica is both MPP and Columnar – best of both worlds

Bank Transaction Data: Teradata vs Vertica

	Teradata	Vertica
Role	System of Record	Analytics Engine
Kind of Data	Final, cleansed, audited, official	Processed, aggregated, enriched
Data Examples	Daily balances, official ledgers, audit trails	Fraud scores, real-time alerts, dashboards
Data Size	10+ years (500B+ rows, PB scale)	30-90 days (50M-1B rows)
Query Speed	Complex joins: 5-10 min	Aggregations: 20-50 ms
Load Speed	Very fast (bulk load)	Fast (columnar insert)
Concurrency	High (1000+ users)	Medium (100+ users)
Example Query	"Official balance for account 12345 as of Dec 31, 2023"	"Flag transactions over $10k from new locations in last 5 min"
Best For	Regulatory reports, end-of-day, audits	Real-time fraud, live dashboards, ad-hoc analytics

Real Bank: 50M transactions/day → Teradata (night) → Vertica (morning for fast queries)

HDFS / Hadoop (Data Lake)

What It Is

- **Type:** Distributed file system
- **Stores:** Raw files (CSV, JSON, logs)
- **Processing:** Spark, MapReduce
- **Schema:** On-read (flexible)

"Dump everything here first, process later"

Example

Examples HDFS, Apache Hadoop, Azure Data Lake Storage, Amazon S3

Best For	Raw data storage, data lakes, big raw data
Who Uses It	Data engineers
Query Speed	Slower (needs Spark engine)

Think: Storage first, process later

Data Flow: Data Lake → Data Warehouse

Raw Data → Data Lake (HDFS) → ETL/ELT → Data Warehouse (MPP) → Dashboards

Quick Takeaway

HDFS stores raw data. MPP (Redshift) analyzes cleaned data very fast using SQL.

Lecture 11: Connection Errors

Common Connection Errors

- **Timeout** - Server not responding
- **Authentication failed** - Wrong credentials
- **Driver not found** - Missing ODBC or JDBC driver
- **Network unreachable** - VPN or Firewall issues
- **Maximum connections reached** - Too many users

Real-Life Connection Issues from Toad World

Issue #1: No Teradata Option

Problem: TDA 2.6 no Teradata in Group

- Root cause: Version too old
- .NET support added in 3.x
- Solution: Upgrade to 3.x or use ODBC

Issue #2: Timeout Error

Problem: Command did not complete within timeout

- Root cause: Long query
- Solution: Increase timeout or optimize query

Why Three Different Databases?

Teradata	Oracle	Vertica
MPP Data Warehouse	OLTP + Warehouse	Columnar Analytics
Big Data (PB scale)	Mixed Workload	Fast Analytics
High Cost	Very High Cost	Moderate Cost
Best for EDW	Best for Transactions	Best for Reporting

No single database fits all needs - choose based on your workload!

Feature Comparison: Architecture

Feature	**Teradata**	**Oracle**	**Vertic**
Architecture	MPP Shared Nothing	Shared Everything	Columnar
Storage Type	Row-based	Row-based	Column
Indexing	Primary Index	B-Tree, Bitmap	Projecti
Compression	Moderate	Good	Excelle

Feature Comparison: Performance

Performance	**Teradata**	**Oracle**	**Vertica**
Big Data Ready	Best	Needs Exadata	Good
Analytics Speed	Fast	Fast (indexed)	Very Fast
Load Speed	Very Fast	Moderate	Fast
Query Concurrency	High	Very High	Medium

Which One is FASTEST?

For Big Data (Billions of rows):

- Teradata #1 - MPP architecture
- Vertica - Very fast for analytics
- Oracle - Needs tuning for big data

For Speed (Queries under 1 second):

- Vertica - Columnar storage
- Oracle - With proper indexes
- Teradata - Good for complex queries

Which Has Big Data Built-in?

- Teradata - Designed for PB scale
- Vertica - Scales well
- Oracle - Needs Exadata for big data

Detailed Comparison Table

Feature	Teradata	Vertica	Oracle
Architecture	MPP Shared Nothing	Columnar MPP	Shared Everything
Best For	EDW, Big Data	Analytics, Reporting	OLTP, Mixed Workload
Compression	Moderate	Excellent	Good
Load Speed	Very Fast	Fast	Moderate
Query Speed (Analytics)	Fast	Very Fast	Fast with indexes
Cost	High	Moderate	Very High
Cloud Ready	Yes (Vantage)	Yes	Yes (OCI)

Use Case: When to Use Which?

Teradata

- Billions+ rows
- Enterprise Data Warehouse
- Complex joins on large tables
- Regulatory compliance needed

Vertica

- Fast analytics on large data
- Time-series analysis
- Log data analysis
- Lower cost than Teradata

Oracle

- OLTP (many writes)
- Existing Oracle ecosystem
- Mixed workloads
- Need PL/SQL

Industry-Specific Recommendations

Banking/Finance

- Teradata - Risk analysis
- Oracle - Core banking
- Vertica - Fraud detection

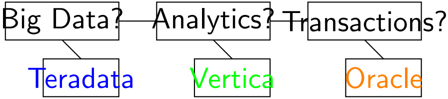
Retail/Ecommerce

- Teradata - Customer 360
- Vertica - Real-time analytics
- Oracle - Order processing

Healthcare

- Oracle - Patient records
- Teradata - Population health
- Vertica - Research analytics

Quick Decision Guide



Simple Rule

- Teradata = Enterprise Big Data Warehouse
- Vertica = Fast Columnar Analytics
- Oracle = Traditional Transactional Database

Real Bank Data Distribution Example

Data Type	Teradata	Vertica	Hive
Transaction History (10+ years)	Primary	Backup	Archive
Customer Master Records	Primary	Copy	Copy
Fraud Detection Alerts	Real-time	Fast Analytics	Batch
Risk Reports (Daily)	Yes	Best	No
Regulatory Filings	Source	Analytics	Staging

Bank's Typical Setup

Teradata = System of Record (Gold Source)
Vertica = Fast Analytics and Reporting
Hive = Data Lake and Archival Storage

Which Data is BIG? Which is FAST?

Big Data (Volume)

Teradata and Hive handle:

- 10+ years transaction history
- Billions of rows per table
- PB scale data warehouses
- Full customer lifecycle data

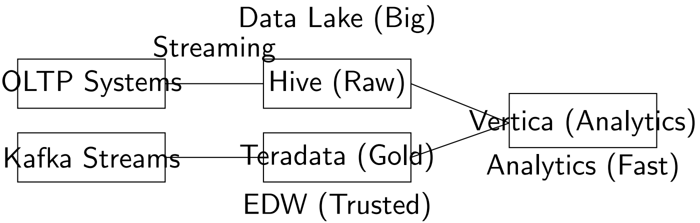
Fast Data (Speed)

Vertica excels at:

- Fraud detection (seconds)
- Real-time risk scoring
- Daily P&L reports
- Customer 360 analytics

Teradata = Big and Safe   Vertica = Fast and Smart   Hive = Cheap and Deep

Bank Example: Daily Data Flow



Example

Daily Volume 10M transactions → Hive (raw) → Teradata (cleansed) → Vertica (aggregated)

Bank Use Case: Fraud Detection

The Problem

Bank needs to detect fraudulent transactions in REAL-TIME

- 50,000 transactions per second
- Must flag within 100ms
- Historical pattern matching needed

The Solution

- **Vertica** - Real-time scoring engine
- **Teradata** - Historical patterns (last 5 years)
- **Hive** - Long-term archive (10+ years)

Query Type	Teradata	Vertica	Hive
Real-time fraud check	50ms	20ms	2000ms
Daily fraud report	5 min	2 min	15 min
Yearly trend analysis	2 hours	30 min	20 min

Bank Data Volume Comparison (Actual Numbers)

Data Type	Volume	Which DB?	Why?
Daily Transactions	50M rows	Vertica	Fast queries
Monthly Aggregates	1.5B rows	Teradata	Reliable EDW
Historical Archive	500B rows	Hive	Low cost
Customer Profiles	10M rows	Teradata	ACID compliance
Fraud Patterns	100M rows	Vertica	Columnar speed

Storage Cost Comparison (per TB)

Teradata: \$5000/month   Vertica: \$1500/month   Hive: \$200/month

Bank: Which Database Wins?

Teradata

Wins when:

- Data must be 100% accurate
- Complex joins needed
- Regulatory reporting
- System of record

Example: End-of-day balances

Vertica

Wins when:

- Speed is critical
- Aggregations needed
- Time-series analysis
- Interactive dashboards

Example: Real-time fraud detection

Hive

Wins when:

- Cost is priority
- Batch processing OK
- Historical analysis
- Data lake storage

Example: 10-year audit archive

Summary: Big vs Fast vs Cheap

	Teradata	Vertica	Hive
Big Data?	✓	✓	✓✓
Fast Queries?	✓	✓✓	X
Real-time?	Moderate	Yes	No
Cheap Storage?	No	No	Yes
ACID Transactions?	✓	X	X
Best For...	Gold Source	Analytics	Archive

Bank's Golden Rule

Teradata = Truth (Single Source of Truth)
Vertica = Speed (Real-time Analytics)
Hive = Scale (Cheap Historical Storage)

The Reality

Queries are long and hard to understand

- Production queries can be 1000+ lines
- Multiple subqueries and CTEs
- Complex business logic
- Performance tuning required

AI for Query Conversion

Converting SQL to PySpark

SQL:

```
SELECT col1, SUM(col2) FROM table GROUP BY col1;
```

PySpark Equivalent:

```
df.groupBy("col1").sum("col2").show()
```

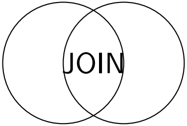
Caution: Always validate AI-generated code!

SQL to PySpark Conversion Examples

SQL	**PySpark**
SELECT * FROM table	df.select("*")
WHERE col = value	df.filter(df.col == value)
GROUP BY col	df.groupBy("col")
ORDER BY col	df.orderBy("col")
JOIN table2 ON key	df.join(df2, "key")

Join Types

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- CROSS JOIN
- SELF JOIN
- ANTI JOIN
- SEMI JOIN



Join Visualization

Join Type	Result
INNER JOIN	Matching rows only
LEFT JOIN	All left + matching right
RIGHT JOIN	All right + matching left
FULL OUTER JOIN	All rows from both
CROSS JOIN	Cartesian product

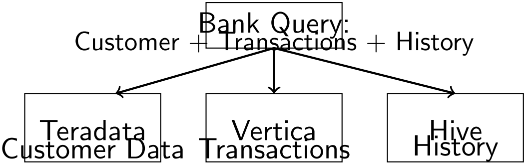
Lecture 18: Query Optimization

Optimization Techniques

1. Use appropriate indexes
2. Filter early (WHERE before JOIN)
3. Avoid SELECT *
4. Use EXPLAIN plan
5. Partition pruning
6. Limit result sets

Query conversion and optimization go hand in hand

The Challenge: Data Spread Across Systems



The Problem

- Customer data in Teradata
- Daily transactions in Vertica
- Historical archive in Hive
- Query needs data from ALL THREE

Solution 1: Federation Layer (Toad Data Point)

Cross-Connection Query in Toad

Toad allows joining tables from different databases in one query.

Example

Toad Cross-Connection Setup

1. Create connection to Teradata (Customer DB)
2. Create connection to Vertica (Transactions DB)
3. Create connection to Hive (History DB)
4. Write SQL that references ALL three

Prerequisite: ODBC drivers for ALL databases must be installed

Complex Bank SQL Query Example

Query: Customer Risk Score with History

```
SELECT c.customer_id, c.customer_name, c.credit_score,
SUM(t.transaction_amount) as total_spend,
COUNT(h.transaction_id) as historical_txns
FROM Teradata.bank_db.customers c
LEFT JOIN Vertica.bank_db.transactions t
ON c.customer_id = t.customer_id
AND t.transaction_date ¿= '2024-01-01'
LEFT JOIN Hive.archive_db.historical_txns h
ON c.customer_id = h.customer_id
WHERE c.credit_score ¡ 650
GROUP BY c.customer_id, c.customer_name, c.credit_score
HAVING SUM(t.transaction_amount) ¿ 10000
```

Note: Syntax varies by tool. Toad uses [DB].table format.

How Banks Actually Handle This (Best Practices)

Method 1: ETL to Central EDW

- Copy all data to Teradata
- Run queries locally
- PRO: Simple, fast joins
- CON: Data duplication

Method 2: Federation (Toad)

- Query across sources live
- No data movement
- PRO: Real-time data
- CON: Slower performance

Method 3: Data Lake

- Land ALL data in Hive
- Use Spark for joins
- PRO: Scalable, cheap
- CON: Batch only

Method 4: API Layer

- Query each source separately
- Join in application code
- PRO: Flexible
- CON: Complex code

Practical Tips for Complex Cross-Database Queries

Step-by-Step Approach

1. Identify source of truth for each data element
2. Start small - test joins with LIMIT 100 first
3. Filter early - push WHERE to each source
4. Use staging tables for frequently joined data
5. Create views to simplify complex logic

Example

Reaching Data Quickly

- Use SELECT TOP or LIMIT for testing
- Check EXPLAIN plan before full execution
- Create materialized views for common joins
- Use batch windows for large historical joins

Toad Syntax Pattern for Cross-Database Queries

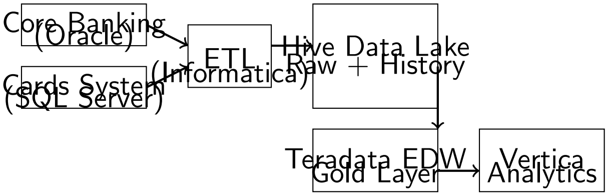
Toad Connection Format

Step 1: Create connections in Toad
- Connection1: Teradata (TD)
- Connection2: Vertica (VC)
- Connection3: Hive (HV)

Step 2: Write query with [Connection].Database.Table

```
SELECT
   TD.customers.cust_id,
   VC.sales.amount,
   HV.history.txn_date
FROM [Teradata].bank_db.customers TD
LEFT JOIN [Vertica].bank_db.sales VC
   ON TD.cust_id = VC.cust_id
LEFT JOIN [Hive].archive_db.history HV
   ON TD.cust_id = HV.cust_id
WHERE VC.sales.amount ¿ 1000
```

Bank's Real-World Data Movement Architecture



Example

Data Flow Sources to ETL to Hive (Landing) to Teradata (Gold) to Vertica (Analytics)

Quick Reference: Reaching Data Across Systems

Scenario	**Best Approach**
Real-time fraud detection	Query Vertica directly (fastest)
End-of-day reporting	Query Teradata (single source of truth)
Historical trend analysis	Query Hive (cheap, scalable)
Join live plus history	ETL history to Teradata first
Ad-hoc analysis across sources	Use Toad Federation
Daily batch reports	Materialized view in EDW

Golden Rule

Do not join across systems in real-time. Move data to one place first.

Definition

SAS Statistical Analysis System - Software suite for advanced analytics, business intelligence, and data management.

SAS Code Example:

```
PROC PRINT DATA=work.sales;
WHERE region = EAST;
RUN;
```

Lecture 20: Why SAS

SAS Strengths

- Industry standard for **Logistic Regression**
- Regulatory compliance (Pharma, Banking)
- Robust data cleaning capabilities
- Powerful data visualization
- Enterprise support

SAS Logistic Regression Example

Logistic Regression Example:

```
PROC LOGISTIC DATA=credit_data;
MODEL default = income age debt;
RUN;
```

Example

Output Includes

- Coefficient estimates
- P-values
- Odds ratios
- Model fit statistics

SAS vs Other Tools

Feature	**SAS**	**Python**	**R**
Regulatory Compliance	High	Low	Low
Learning Curve	Steep	Moderate	Steep
Cost	High	Free	Free
Visualization	Good	Excellent	Good
Big Data Support	Moderate	Excellent	Moderate

Building ML Model for Wire Transfer Fraud: Where to Get Data

Where Does Each Data Type Live in the Bank?

Data Needed for Model	Source Database	Why This DB?	Access Method
Transaction Amount, Time, Date	Oracle Exadata	Core OLTP banking ledger	JDBC / ODBC
Sender Account History (last 30 days)	Teradata	EDW historical aggregation	SQL
Receiver Account History	Teradata	EDW multi-year ledger view	SQL
Customer Profile (Age, Location, Tenure)	Teradata	Single source of truth (EDW)	SQL / JDBC
Past Fraud Flags (Labels for Training)	Teradata	Audited compliance records	SQL
Device IP, Location, Session Data	Vertica	Columnar = fast log analytics	ODBC / JDBC
Historical Wire Transfers (5+ years)	Hive	Cheap archival storage	HiveQL / Spark
Geographic Risk Scores	Teradata	Central reference data	SQL
Counterparty Risk Rating	Oracle Exadata	Risk management system	ODP.NET / SQL

How to Get ALL Data for Modeling

- **Step 1:** Extract from Exadata (core transaction) + Teradata (labels + profiles)
- **Step 2:** Join with Vertica (session logs) + Hive (historical archives)
- **Step 3:** Build feature store in Vertica or Spark Feature Store
- **Step 4:** Train model using PySpark or Python (pandas/scikit-learn)

Loan Default Model: Where Consumer Data Lives in Bank

Consumer Loan Default Risk - Data Sources

Consumer Data Element	Source DB	Why This DB?	Update Frequency
Credit Score (Bureau Data)	Teradata	System of record for risk	Monthly
Income / Employment History	Teradata	Master profile (EDW)	Quarterly
Existing Loan Balances	Oracle Exadata	Core loan accounting platform	Real-time / Daily
Payment History (12 months)	Teradata	Consolidated ledger view	Daily
Late Payment Flags	Oracle Exadata	Loan servicing engine (ACID)	Real-time
Debt-to-Income Ratio	Oracle Exadata	Calculated at origination	Static per application
Customer Address / City	Teradata	Master record (MDM Hub)	As needed
Economic Data (Unemployment by City)	Hive	External batch data staging	Monthly
Property Value (Collateral)	Oracle Exadata	Loan origination platform	Yearly / At Appraisal
Past Default Labels (Training)	Teradata	Historical risk warehouse	Static
Transaction Behavior (Spending patterns)	Vertica	Columnar analytics on card logs	Daily batch / Streaming

Different Data = Different Databases

- Teradata = Customer Master, Historical Payment Views, Credit Scores, Default Labels
- Oracle Exadata = Real-time Balances, Late Flags, DTI, Origination/Collateral Specs

Congratulations!

You've completed the course.

What You Learned

- Data Permissions and Governance
- Connection Methods (ODBC, Native, SSO)
- Teradata, Oracle, Vertica differences

Course Summary - Part 2

Skills Acquired

- Query Optimization Techniques
- SQL to PySpark Conversion
- SAS Analytics Basics
- Troubleshooting Real Issues

All Learning Objectives Completed!

Key Takeaways

1. Always check permissions before querying
2. Choose the right database for your workload
3. Native drivers are better than ODBC when available
4. Vertica requires 64-bit ODBC driver
5. Windows SSO works for Teradata and Oracle only
6. Always validate AI-generated code

Next Steps

Continue Your Learning

- Practice with real datasets
- Try the exercises in each section
- Join the discussion forum
- Rate this course on Udemy

Thank you for enrolling!

Keep Learning, Keep Growing!